

Cost Splitting

Hands-On 1 Problem (contributed by J. C. Santillana)

Whenever a friend or loved one has a celebration coming up, some people would suggest to split the cost (gift and food) among those who agree to the idea. Sometimes the computation can get messy so a program to help them know the exact amount would be helpful. For this problem, a programmer has started developing the program but has yet to finish it. Your task is to complete the missing parts to complete the program.

Assumptions:

- Assume all inputs are valid (ie. no negative inputs)
- The number of people backing out cannot be more than the number of people in the group
- The maximum amount for the gift cannot exceed 9999.00
- All displayed monetary values must be limited to 2 decimal places
- The number of people backing out can be zero(0)

The user will be asked to input the following information:

- The number of people in the group
- The price of the gift
- The number of people who backed out

The program will take the inputs and display the amount each person will contribute for the gift. If the number of people remaining to contribute are 7 or more, it is assumed that they will all also surprise the celebrant with food split amongst them not including the celebrant. For the food total, service fee and delivery fee will be added.

The user will then be asked to input some additional information:

- Ask how much for the price of the food (with 12% VAT inclusive)
- Ask if there are any PWD in the group
 - This will affect the final cost since there will now be VAT discount(12%) and PWD discount(20%) to be taken into account for the computation

There are 2 files in this file:

- Cost-main.c – contains the main() function
- Cost-lastname.c – file where you will place your solutions

Tasks:

- Task 1: Implement the costPerPerson() function (5 pts)
This function will compute and return the cost of the gift per person.
Example: If the gift costs 2000 and there are 4 people in the group the function will return 500.

- Task 2: Implement the `newCostPerPeson()` function (10 pts)
This function will compute the new price per person less the number of backouts and update the cost of the gift per person.

- Task 3: Implement `computeFood()` function (15 pts)

This function will compute and return the overall cost of the food which includes the 10% service fee and 150 pesos delivery fee.

Note:

- The number of people at this point is whoever is left after accounting for those who backed out. It is assumed that those who backed out will no longer join for the food either.
- The 10% service fee is only for the total price of the food cost which DOES NOT INCLUDE the delivery fee.
- You will need to compute for the following in sequence:
 1. Food price without 12% VAT = Price of food / (1 + VAT rate)
 2. VAT exempt amount = Price of food - Price of food without VAT
 3. PWD (20%) discount = Price of food without VAT * 0.2
 4. Discount total = VAT exempt amount + PWD discount amount
 5. Total amount of the food with discount = Food price - discount total
 6. Service fee = Total amount of the food with discount * 0.1
 7. Overall total = The sum of Total amount of the food with discount + computed service fee + delivery fee

Example:	
Price of the food = 3700	
Food price without 12% VAT:	$3700 / (1 + 0.12) = 3303.57$
VAT exempt amount:	$3700 - 3303.57 = 396.43$
PWD discount:	$3303.57 \times 0.2 = 660.71$
Discount total:	$396.43 + 660.71 = 1057.14$
Total of the food w/distance:	$3700 - 1057.14 = 2642.86$
Service fee:	$2642.86 \times 0.1 = 264.29$
Overall total:	$2642.86 + 264.29 + 150 = 3057.15$
3057.15 should be returned by the function	

The above will be needed by the foodBreakdown() function call which is part of this function.

- Task 4: Complete the foodBreakdown function() (10 pts)
This function will display the breakdown for the food cost. It is already partially defined. Your task is to add the missing parts.
- Task 5: Complete the giftBreakdown() function (5 pts)
This function will display the breakdown of the gift cost. It is already partially defined. Your task is to add the missing parts. This function will also call the newCostPerPerson() function.

Sample runs:

Scenario 1: 7 people, 5000 gift price, 2 backout

```
Enter no. of people in the group: 7
Enter total price of the gift: 5000
Number of people who backed out: 2
-----

Gift Cost Breakdown:
-----
Total price of gift with 12% VAT:                5000.00
Initial Cost of gift per person (7 people):      714.29
-----
Final cost of gift per person (5 people):        1000.00
```

Scenario 2: 5 people, 4000 gift price, 0 backout

```
Enter no. of people in the group: 5
Enter total price of the gift: 4000
Number of people who backed out: 0
-----

Gift Cost Breakdown:
-----
Total price of gift with 12% VAT:                4000.00
Initial Cost of gift per person (5 people):      800.00
-----
Final cost of gift per person (5 people):        800.00
```

Scenario 3: 12 people, 5400 gift price, 3 backout, 2500 food price, no pwd

How much for the food? 2500

Are there any PWD in the group?(y/n) n

Gift Cost Breakdown:

```
-----  
Total price of gift with 12% VAT:                5400.00  
Initial Cost of gift per person (12 people):      450.00  
-----  
Final cost of gift per person (9 people):         600.00
```

Food Cost Breakdown:

```
-----  
VAT Discount 12%:                0.000  
PWD Discount 20%:                0.000  
Total Discount:                  0.000  
Sub-Total:                       2500.000  
Service Fee:                     250.000  
Delivery Fee:                    150.000  
-----  
Total:                           2900.000  
Cost of food per person:         322.22  
-----  
Overall total per person:        922.22  
-----
```

Scenario 4: 11 people, 3000 gift price, 2 backout, 3700 food price, with pwd

```
Enter no. of people in the group: 11
Enter total price of the gift: 3000
Number of people who backed out: 2
-----

How much for the food? 3700
Are there any PWD in the group?(y/n) y

Gift Cost Breakdown:
-----
Total price of gift with 12% VAT:                3000.00
Initial Cost of gift per person (11 people):      272.73
-----
Final cost of gift per person (9 people):          333.33

Food Cost Breakdown:
-----
VAT Discount 12%:                396.428
PWD Discount 20%:                660.714
Total Discount:                  1057.143
Sub-Total:                       2642.857
Service Fee:                     264.286
Delivery Fee:                    150.000
-----
Total:                           3057.143
Cost of food per person:          339.68
-----
Overall total per person:        673.02
-----
```

To debug and test your program with the main() in a separate file, you need to do the following:

1. You need to have all your files in the same folder (directory).
2. Then, use the following instruction to compile: `gcc -Wall -std=c99 Cost-main.c`
This will produce a.exe. Note that there is no need to compile Cost-LASTNAME.c separately.
3. Run in the command prompt as usual. For example, given a.exe generated from step 2, just type a in the command prompt.

DELIVERABLES: Your C program source file Cost-LASTNAME.c with your own last name as filename. For example, if your lastname is SANTOS, then the source file should be named as Cost-SANTOS.c. Upload your source file in AnimoSpace before the indicated submission time.

TESTING & SCORING:

- Your program will be compiled via gcc -Wall -std=c99. Thus, for each function that does not compile successfully, the score for that function is 0.
- Your program will be tested by your instructor with other sample inputs (different from the ones given to you) and with function calls of different parameter values. It is also possible for the instructor to test your cost.c file (the functions) using a different main() program.
- Full credit will be given for the function only if the student's implementation are all correct for all the test values used by the instructor during checking AND only if the student's implementation complied with the requirement and did not violate restrictions. Deductions will be given if not all test cases produce correct results OR if restrictions were not followed.